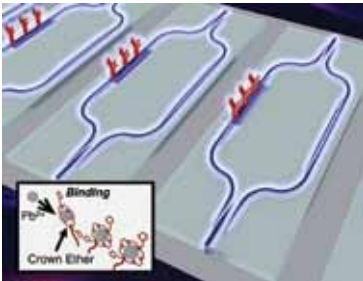




compromise productivity. Tests on a 3-axis CNC machine and a 3D printer showed production cycle times reduced by 38% and 17%, respectively. The digital twin model, incorporating physics-based principles and real-time sensor data, predicts and compensates for variables, managing uncertainty by adjusting speed to maintain quality.

Affordable sensor for lead contamination

Engineers from MIT, Nanyang Technological University, and various corporations have created an affordable device to detect and measure lead in water, addressing a significant



Artist's impression of the chip surface, showing the on-chip light interferometer used to sense the presence of lead. The lead binding process to the crown ether is shown in the inset (Credit: Jia Xu Brian Sia)

global health issue. Traditional methods are complex and costly, while simpler tests lack precision. US standards require less than 15 parts per billion (ppb) of lead in drinking water. The new system detects lead concentrations as low as 1 ppb with high accuracy using a portable device with a chip-based

sensor. This technology employs photonic chips and crown ethers to capture lead ions, achieving near-instant results. Market release is anticipated within two to three years.

Flexible electronic device for spinal injury treatment

University of Cambridge team has developed a flexible electronic device that wraps around the spinal cord, potentially revolutionising spinal injury treatment. This device, created by engineers, neuroscientists, and surgeons, records and stimulates nerve signals, providing a safer, non-invasive alternative to current high-risk surgeries. Traditional methods involve electrodes piercing the spinal cord or brain implants,



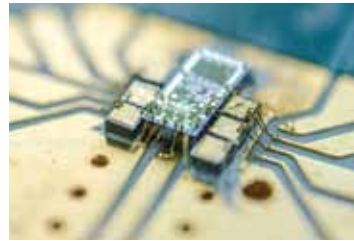
An intraoperative image of the device wrapped around the longitudinal axis of the spinal cord (Credit: Science Advances)

but the new device offers 360-degree recording, capturing a complete picture of spinal cord activity. The biocompatible device, made using advanced photolithography and thin-film deposition techniques, demonstrated low-latency stimulation in animal

and human cadaver tests. Although treatments are years away, this innovation could soon aid in monitoring spinal cord activity during surgeries and improve treatments for conditions like chronic pain and hypertension.

World's tiniest quantum light detector embedded on silicon chip

University of Bristol researchers have made a significant advancement in quantum technology by integrating a tiny quantum light detector onto a silicon chip. The detec-



The silicon ePIC quantum chip, mounted on a printed circuit board for testing and similar to a motherboard inside a personal computer (Credit: University of Bristol)

tor, thinner than a human hair, enables high-speed quantum communications and optical quantum computing. The 80x220 micrometre circuit uses established fabrication techniques, enhancing prospects for integration into sensing and communication

technologies. The new electronic-photonic integrated chip increases detector speed by ten times and reduces the footprint by fifty times. Future research aims to integrate more quantum technology hardware at the chip scale and improve detector efficiency for various applications.

Skin like material enhances connectivity for wearables

Researchers from Rice University and Hanyang University have developed a new material that mimics skin's elasticity while maintaining signal strength in electronics, enabling next-generation wearable devices with continuous, wireless, battery-free functionality. The material, made by embedding highly dielectric ceramic nanoparticles into an elastic polymer, counteracts motion's disruptive effects on electronics. This minimises energy loss and dissipates heat, stabilising radio-frequency communication. Applications include wearable medical devices, soft robotics, and high-performance antennas. Testing on various stretchable wireless devices demonstrated stable performance under strain, supporting communication up to 30 metres.

AI chips could get sense of time

Memristors can enhance artificial neural networks (ANNs) to process time-sensitive data like audio and video more efficiently. The University of Michigan introduces the first tuneable 'relaxation time' memristor, improving processing ca-